Theorem (Closure Under Union)

The set of regular languages is closed under union.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 \cup L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two DFAs

$$D_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$D_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.

We construct a new DFA

$$D = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 \cup L_2$, where:

1. $Q = Q_1 \times Q_2$.
2. Σ is the common input alphabet of D_1 and D_2.
3. The transition function δ is defined as

   $$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

   for all $r_1 \in Q_1$, $r_2 \in Q_2$, and $a \in \Sigma$.
4. $q_0 = (q_1, q_2)$.
5. The set of accepting states is

   $$F = \{(r_1, r_2) \mid r_1 \in F_1 \text{ or } r_2 \in F_2\}.$$

   Equivalently,

   $$F = (F_1 \times Q_2) \cup (Q_1 \times F_2).$$

The DFA D simulates both D_1 and D_2 simultaneously and accepts a string if either D_1 or D_2 accepts it.

Therefore,

$$L(D) = L_1 \cup L_2.$$

Since D is a DFA that recognizes $L_1 \cup L_2$, the language $L_1 \cup L_2$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under union.}}$$

□

Theorem (Closure Under Complement)

The set of regular languages is closed under complement.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L \in \mathcal{R}_L, \qquad \overline{L} \in \mathcal{R}_L.$$

Since L is regular, there exists a DFA

$$M = (Q, \Sigma, \delta, q_0, F)$$

that recognizes L.

We construct a new DFA

$$M' = (Q', \Sigma', \delta', q_0', F')$$

that recognizes $\overline{L}$, where:

1. $Q' = Q$.
2. $\Sigma' = \Sigma$.
3. $\delta'(q, a) = \delta(q, a)$, for all $q \in Q$ and $a \in \Sigma$.
4. $q_0' = q_0$.
5. $F' = Q - F$.

The DFA M' has the same states, alphabet, transitions, and start state as M, but its accepting and non-accepting states are interchanged.

Therefore,

$$L(M') = \overline{L} = \Sigma^* - L.$$

Since M' is a DFA that recognizes $\overline{L}$, the language $\overline{L}$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under complement.}}$$

□

Theorem (Closure Under Intersection)

The set of regular languages is closed under intersection.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 \cap L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two DFAs

$$D_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$D_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.

We construct a new DFA

$$D = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 \cap L_2$, where:

1. $Q = Q_1 \times Q_2$.
2. Σ is the common input alphabet of D_1 and D_2.
3. The transition function δ is defined as

   $$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

   for all $r_1 \in Q_1$, $r_2 \in Q_2$, and $a \in \Sigma$.
4. $q_0 = (q_1, q_2)$.
5. The set of accepting states is

   $$F = \{(r_1, r_2) \mid r_1 \in F_1 \text{ and } r_2 \in F_2\}.$$

   Equivalently,

   $$F = F_1 \times F_2.$$

The DFA D simulates both D_1 and D_2 simultaneously and accepts a string if both D_1 and D_2 accept it.

Therefore,

$$L(D) = L_1 \cap L_2.$$

Since D is a DFA that recognizes $L_1 \cap L_2$, the language $L_1 \cap L_2$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under intersection.}}$$

□

Theorem (Closure Under Difference)

The set of regular languages is closed under difference.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 - L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two DFAs

$$D_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$D_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.

We construct a new DFA

$$D = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 - L_2$, where:

1. $Q = Q_1 \times Q_2$.
2. Σ is the common input alphabet of D_1 and D_2.
3. The transition function δ is defined as

$$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

   for all $r_1 \in Q_1$, $r_2 \in Q_2$, and $a \in \Sigma$.
4. $q_0 = (q_1, q_2)$.
5. The set of accepting states is

$$F = \{(r_1, r_2) \mid r_1 \in F_1 \text{ and } r_2 \notin F_2\}.$$

   Equivalently,

$$F = F_1 \times (Q_2 - F_2).$$

The DFA D simulates both D_1 and D_2 simultaneously and accepts a string if D_1 accepts it and D_2 rejects it.

Therefore,

$$L(D) = L_1 - L_2.$$

Since D is a DFA that recognizes $L_1 - L_2$, the language $L_1 - L_2$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under difference.}}$$

□